

CSE 311:

Data Communication

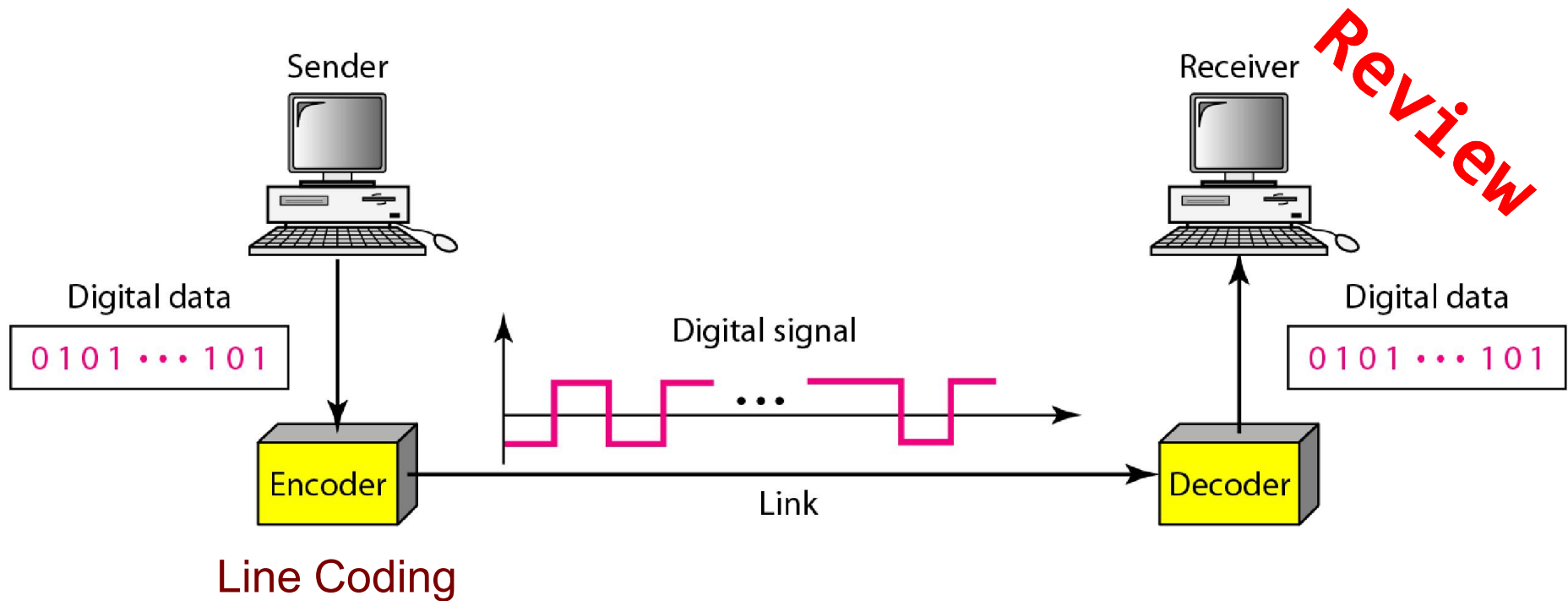
Instructor:

Dr. Md. Monirul Islam

Digital-to-Digital Conversion: **Line Coding, Block Coding and** **Scrambling**

From
Data Communications and Networking, 5th Edition
By Behrouz A. Forouzan

Line Coding and Decoding



- converts *digital data* elements to *digital signal* elements

- Example:

$1 \rightarrow +V$

$0 \rightarrow -V$

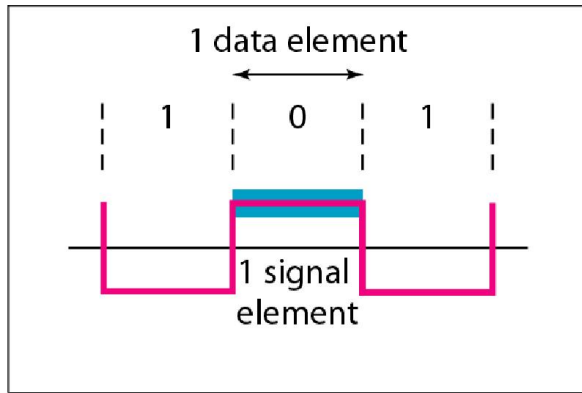
Data Element and Signal Element

Review

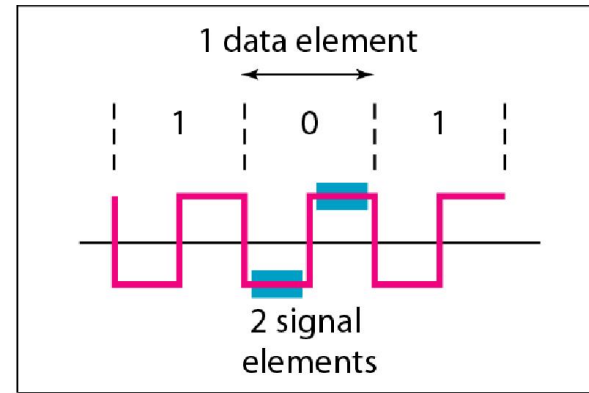
- data element
 - smallest entity to represent a piece of information
 - 0, 1, etc,
- signal element
 - shortest unit of a digital signal
 - Each represents one or more data element
- signal element is carrier, data element is carried
- r = No. of data element is carried by each signal element

Data Element and Signal Element

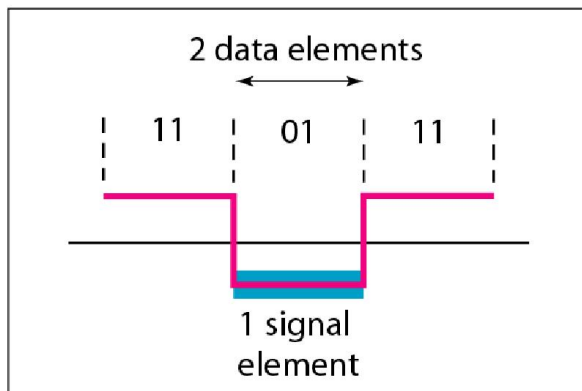
Review



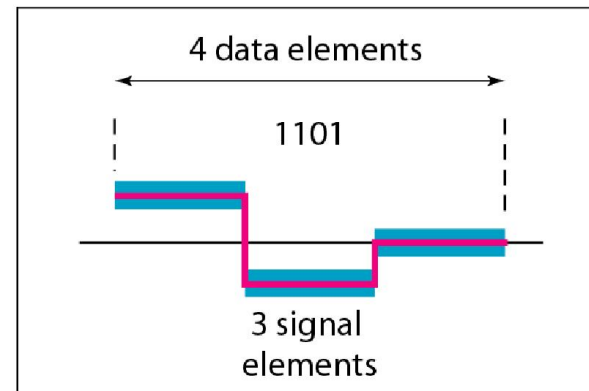
a. One data element per one signal element ($r = 1$)



b. One data element per two signal elements ($r = \frac{1}{2}$)



c. Two data elements per one signal element ($r = 2$)



d. Four data elements per three signal elements ($r = \frac{4}{3}$)

Data Rate and Signal Rate

Review

- data rate, N
 - No. of **data** elements sent in 1 second
 - Other names: bit rate
- signal rate, S
 - No. of **signal** elements sent in 1 second
 - Other names: baud rate, pulse rate, modulation rate
- $r = N/S$ alternately, $S = N/r$

Line Coding Guidelines

Review

- r should be higher, $S = N/r$ should be lower
- Minimize *Baseline wandering*:
 - Running average of incoming signal power is *baseline*
 - Baseline is compared with incoming signal for decoding
 - Long runs of 0 or 1 drifts the baseline

Line Coding Guidelines

Review

- Minimize *DC components*:
 - Constant voltage for long period
 - increaes low frequency components
 - bandpass channels **do NOT** support low frequencies

Line Coding Guidelines

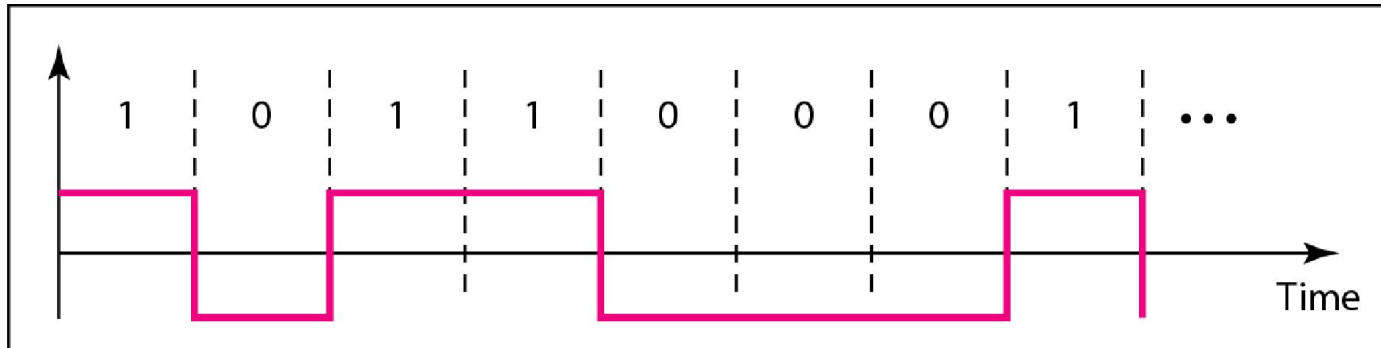
Review

- Self synchronization
 - Sender's and receiver's clock must have same bit interval
 - Different intervals may misinterpret incoming signal

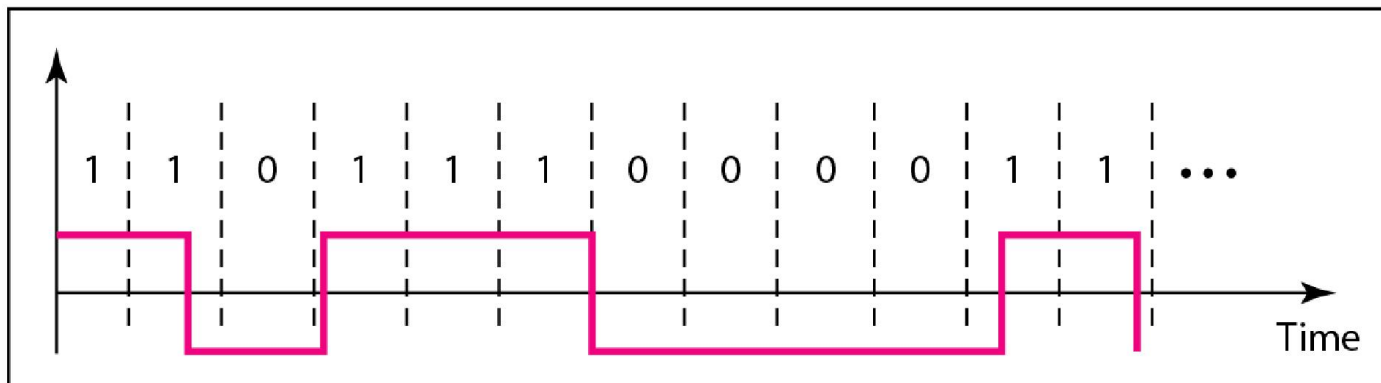
Line Coding Guidelines

Review

Illustration of lack of synchronization



a. Sent



b. Received

Line Coding Guidelines

Review

- Built-in Error Detection
 - Error occurs during transmission
 - Must be able to detect and/or correct them
 - Example: certain type of signal transition (change) is NOT part of the coding

Line Coding Guidelines

Review

- Higher Noise immunity
 - certain line coding prevents noises to be added to the signal
 - It is better than error detection

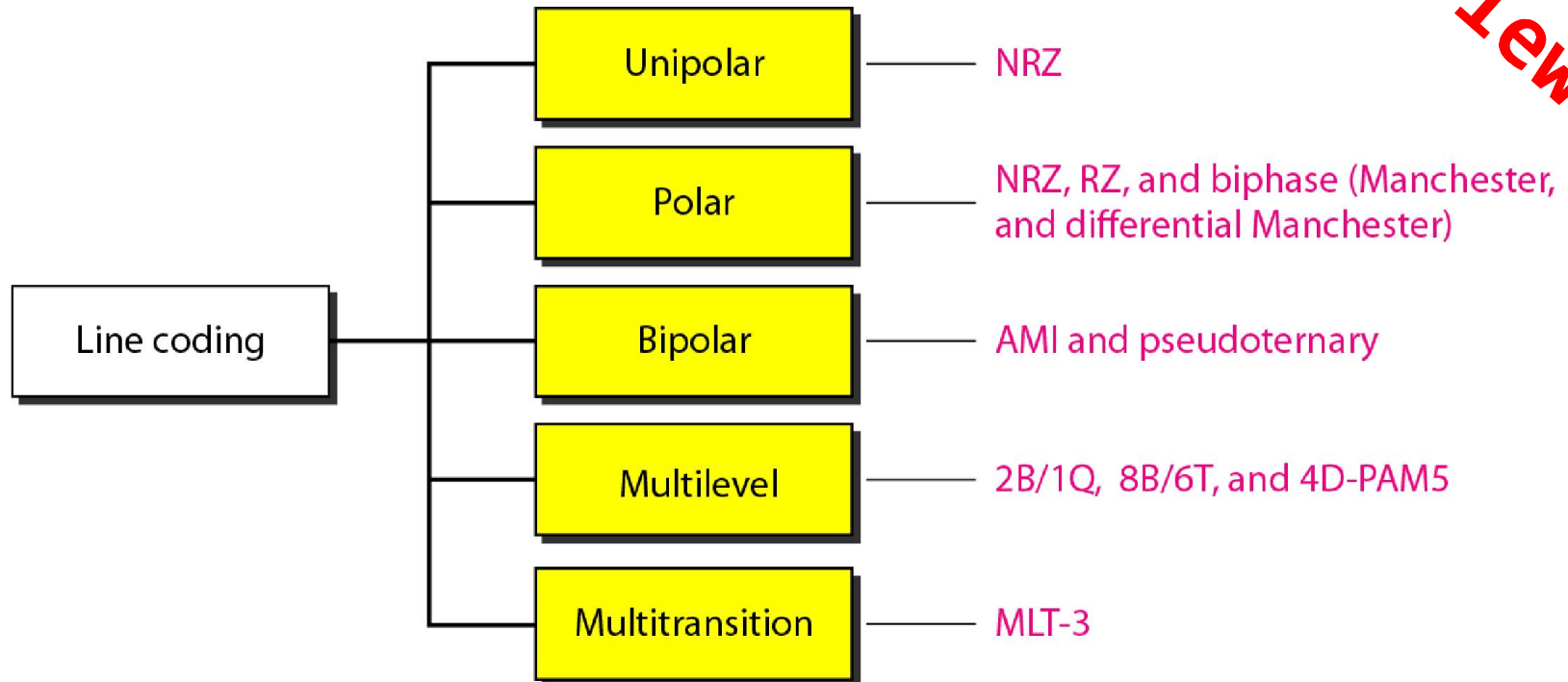
Line Coding Guidelines

Review

- Complexity
 - Simpler is better
 - A coding with FOUR signal levels is more difficult to interpret than one that uses TWO signal levels

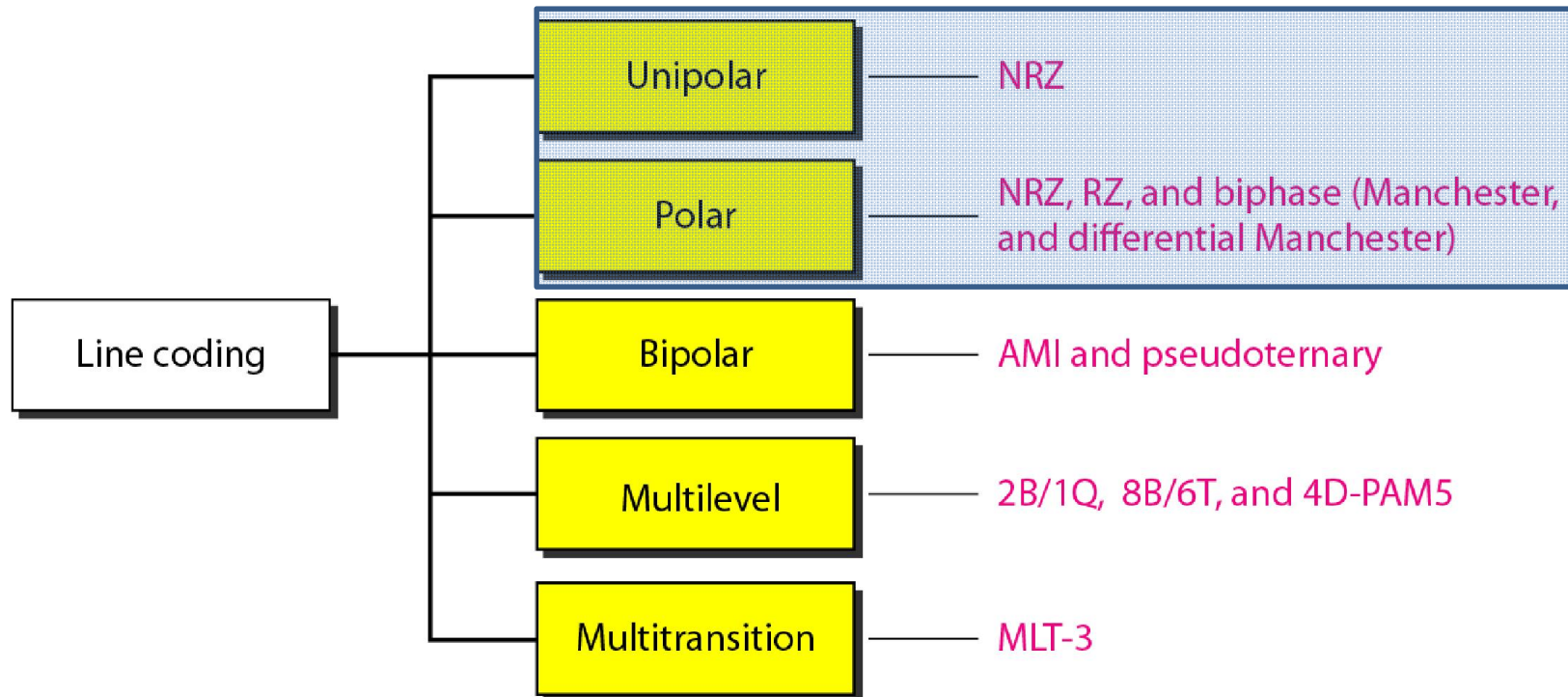
Line Coding Schemes

Review



Line Coding Schemes

Review



Polar: NRZ-L and NRZ-I

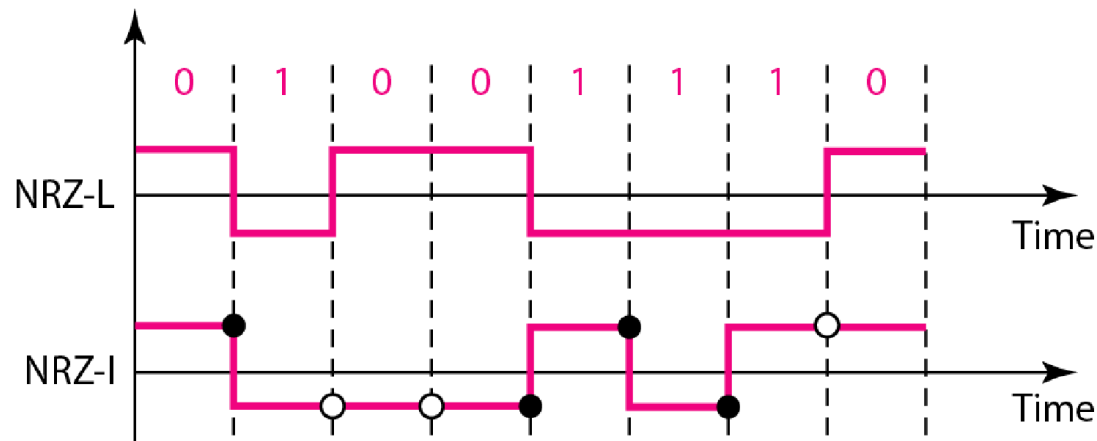
Review

- Polar NRZ-L

$+V$: 0 and $-V$: 1

- Polar NRZ-I

Inversion: next bit is 1, **No Inversion:** next bit 0

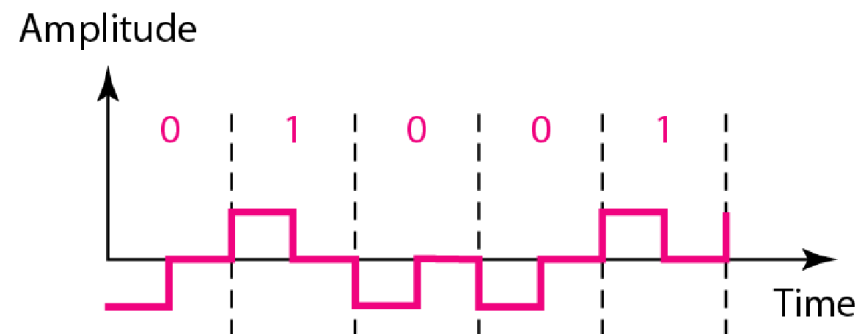


Polar: Return to Zero (RZ)

- 3 signal levels: $+V$, $-V$, 0
- Signal transition in the middle of the bit

– bit 0: $-V$ to 0 bit 1: $+V$ to 0

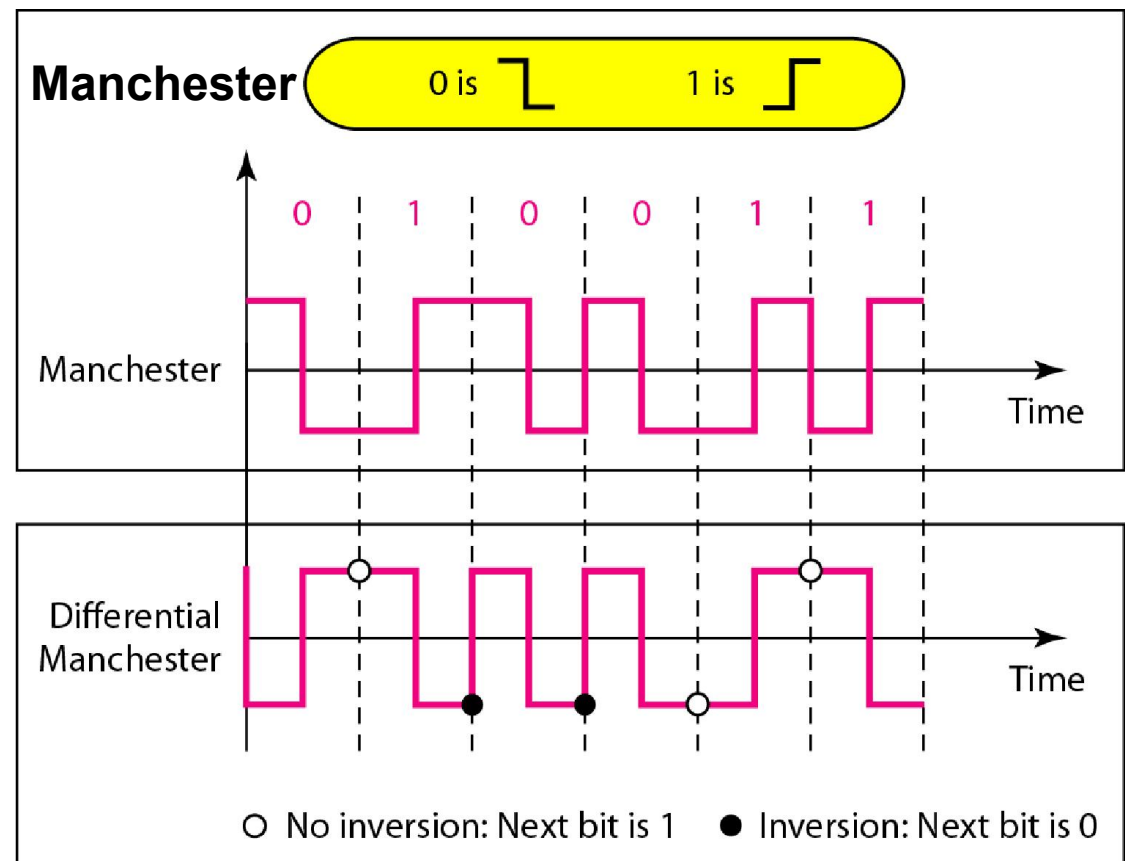
Review



Polar: Biphase-Manchester and Differential Manchester

Review

- Manchester
 - Level determines bits
- Differential Manchester
 - Inversion/no inversion determines bits



Bipolar

- 3 signal levels: 0, +V, -V
- 2 variations: alternate mark inversion (AMI) and Pseudoternary
- AMI
 - bit 0: signal level 0
 - bit 1: successively alternates between +V and -V
- Pseudoternary
 - bit 1: signal level 0
 - bit 0: successively alternates between +V and -V

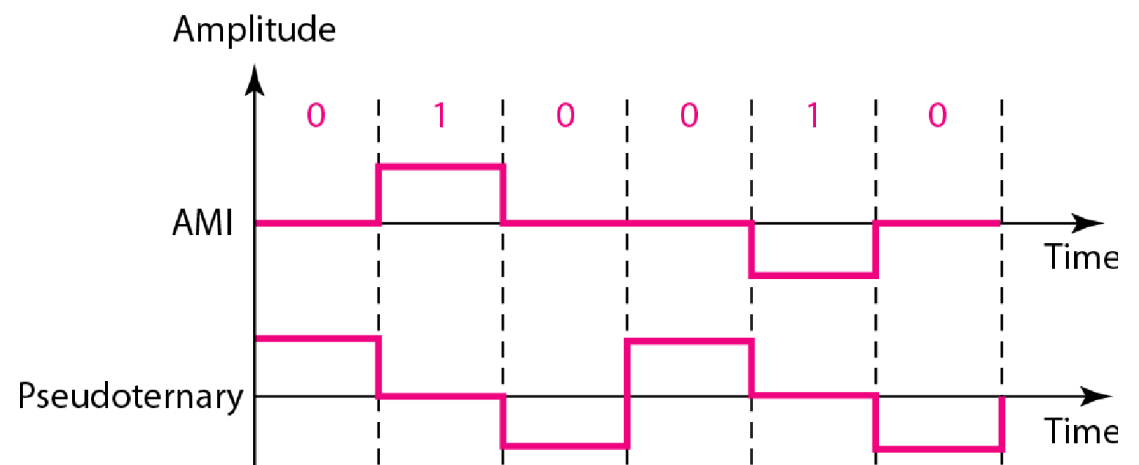
Bipolar

- AMI

- bit 0: signal level 0
- bit 1: successively alternates between $+V$ and $-V$

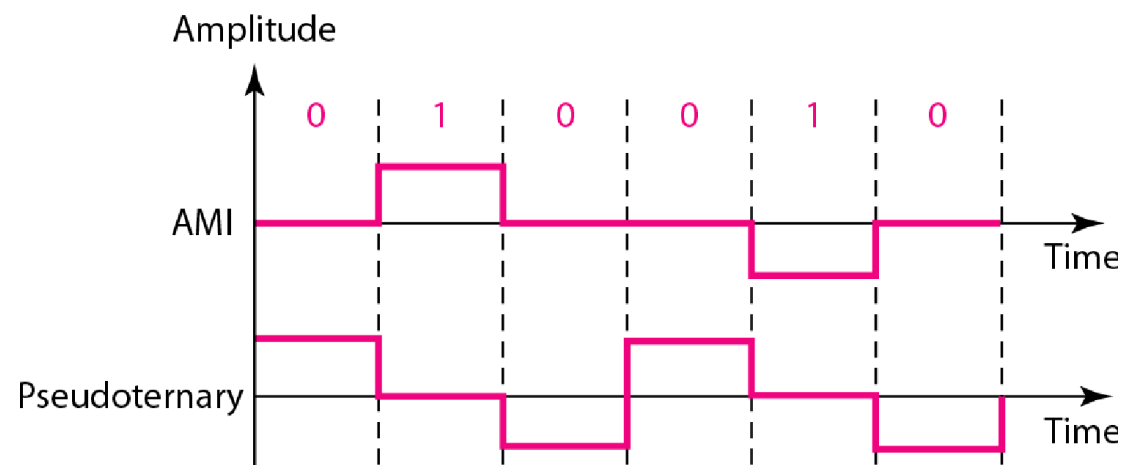
- Pseudoternary

- bit 1: signal level 0
- bit 0: successively alternates between $+V$ and $-V$



Bipolar

- Pros
 - $r = 1$
- Cons
 - 3 signal elements
 - NO synchronization
 - NO error detection



Multilevel Schemes

- Target: to increase r or *bit rate*
- Individual bit is NOT coded
- rather a sequence of m bits is coded by a sequence of n signal elements
- Assuming, No. of signal levels is L
 - Possible data patterns: 2^m
 - Possible signal patterns: L^n

2^m data patterns $\rightarrow L^n$ signal patterns

Multilevel Schemes

2^m data patterns $\rightarrow L^n$ signal patterns

- $2^m > L^n$: mapping is NOT possible
- $2^m = L^n$: exact mapping is possible
- $2^m < L^n$:
 - Redundant signal patterns
 - Very flexible mapping is possible
 - Better noise immunity and error detection

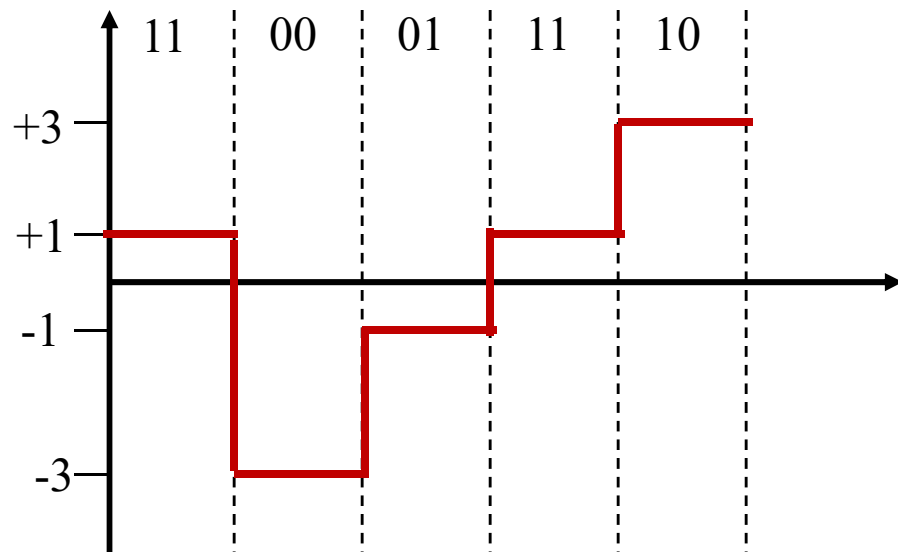
Multilevel Schemes

- Coding symbol: short representation is $mBnL$
- L is replace by character code
 - $L = 2 : B$ meaning Binary
 - $L = 3 : T$ meaning Ternary
 - $L = 4 : Q$ meaning Quaternary

$2B1Q$ Scheme

Rules

00: -3
01: -1
10: +3
11: +1

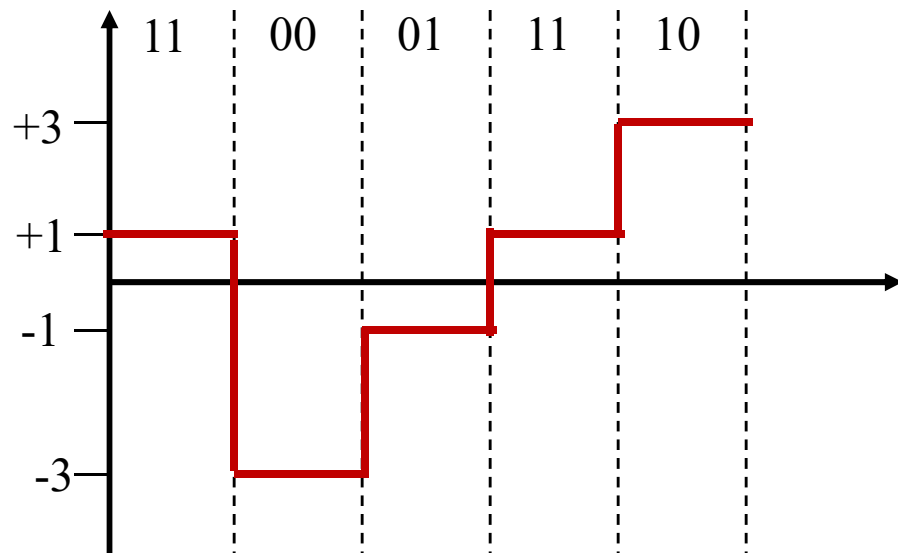


Multilevel Schemes

- Pros
 - $r = 2$
 - Used in DSL to provide high speed connection to the Internet
- Cons
 - 4 signal elements,
 - Baseline wandering, DC components are possible
 - NO redundancy, NO error detection

2B1Q Scheme
Rules

00: -3
01: -1
10: +3
11: +1



Multilevel Schemes: 8B6T

- $2^8 = 256$ different data patterns
- $3^6 = 729$ different signal patterns of 3 levels, +V, -V and 0
- No. of redundant signal patterns $729 - 256 = 473$
- Flexible pattern mapping
- Synchronization and error detection by 473 redundant signal patterns

Multilevel Schemes: 8B6T

- All mapped signal patterns have weight +1 or 0 dc values

Mapping Table		
Data patterns	Signal patterns	weight
00010001	− 0 − 0 + +	0
01010011	− + − + + 0	+1
01010000	+ − − + 0 +	+1

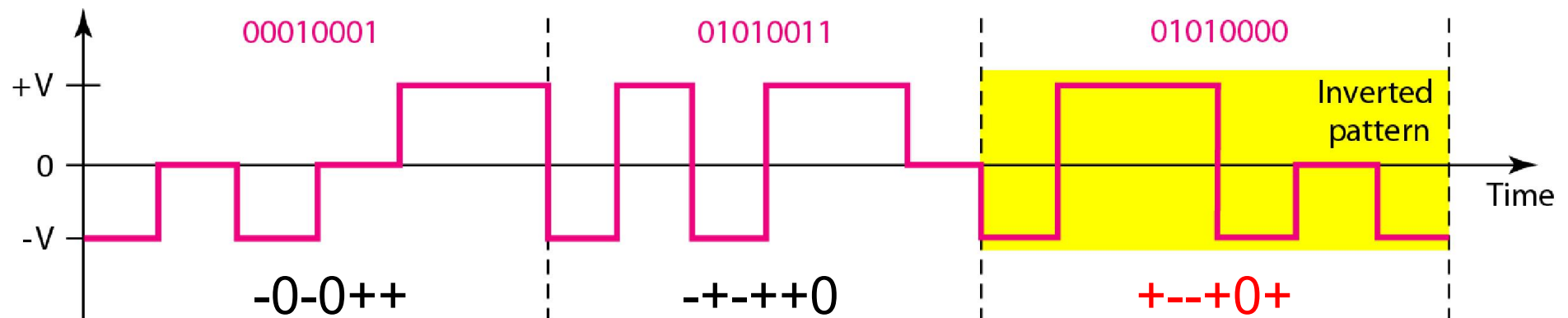
- NO signal patterns have weight -1
- DC balance is maintained by inverting **polarity** of signal pattern

if necessary + − − + 0 + is replaced by − + + − 0 − (WEIGHT: -1)

Multilevel Schemes: 8B6T

Let, we have to send data sequence

000100010101001101010000



Weight: 0

Weight: +1

Weight was +1

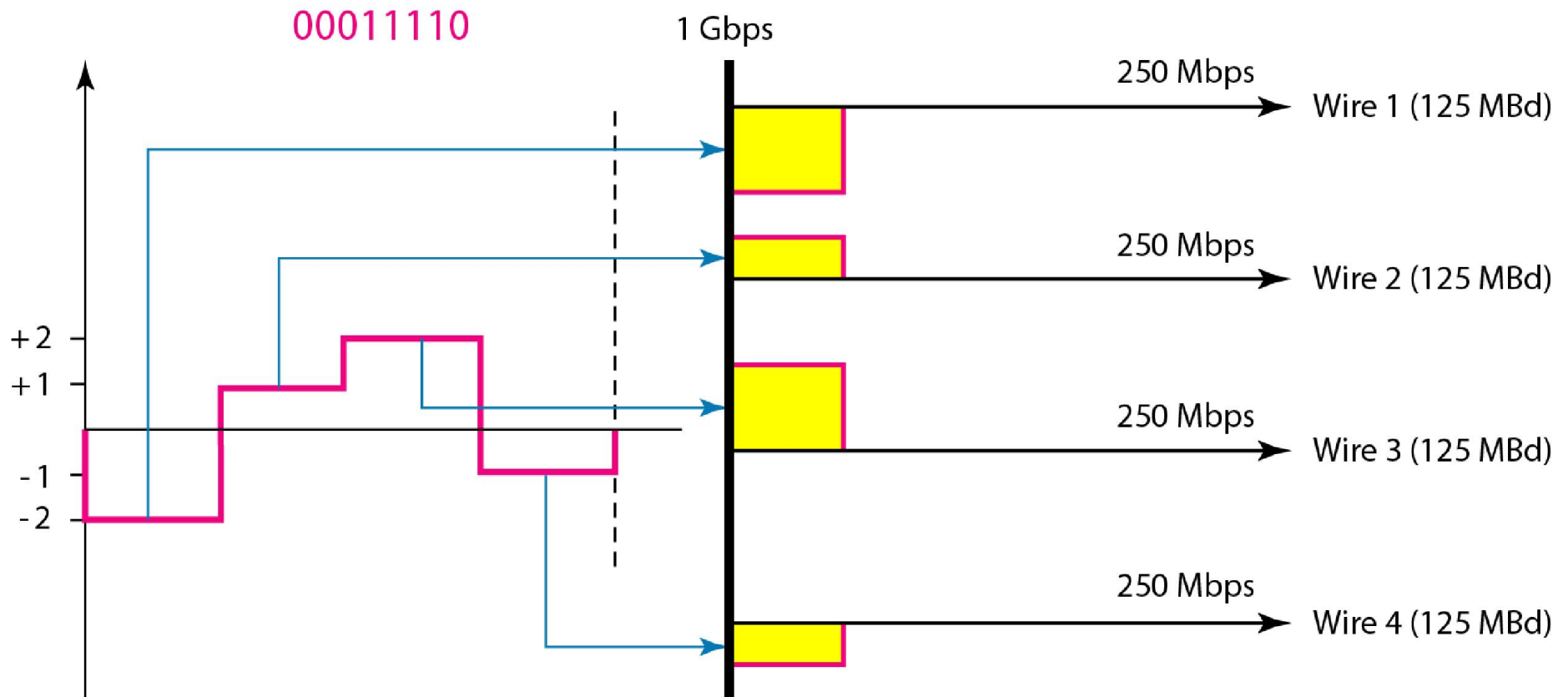
Now it is -1

to ensure DC balance

Multilevel Schemes: *4D-PAM5*

- 4 *D*imensional: 4 signal elements, each sent through separate wire
- *P*ulse *A*mplitude *M*odulation 5: 5 different levels -2, -1, 0, +1, +2
- Used signal levels: -2, -1, +1, +2
- Total signal patterns: $4^4 = 256$, which support 256 binary data patterns
- *4D-PAM5* is equivalent to *linear 8B4Q*

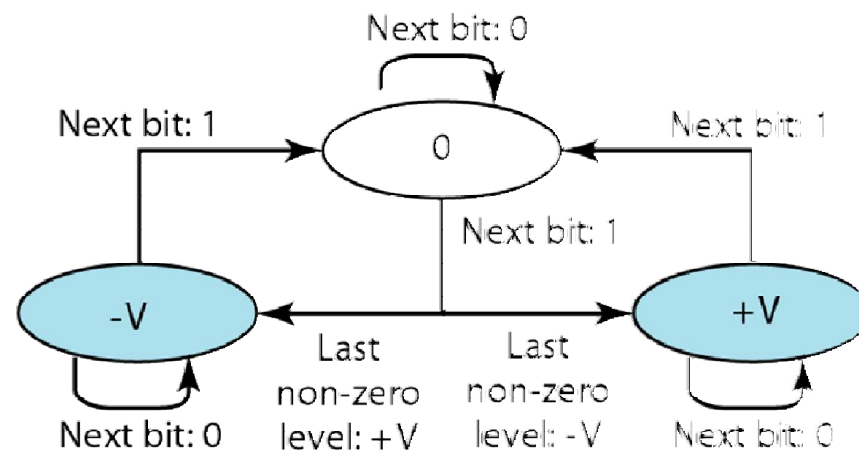
Multilevel Schemes: *4D-PAM5*



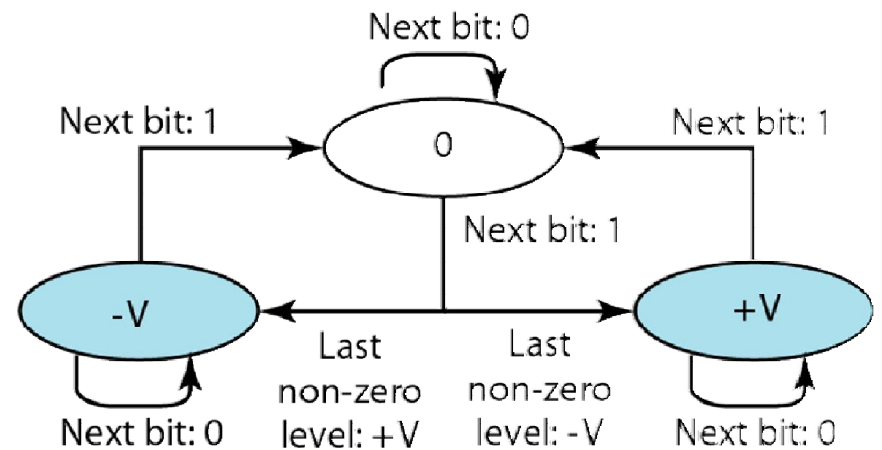
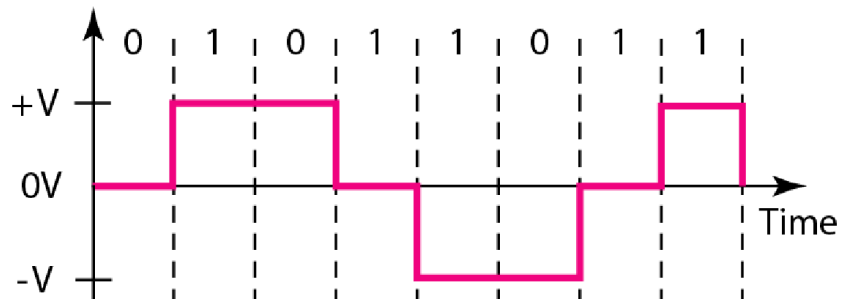
- 1 Gbps Gigabit LAN through 4 copper wires of 125 Mbaud each
- Self synchronization, NO DC components

Multitransition: MLT-3

- Similar to differential coding like NRZ-I and differential Manchester
- Uses more than 2 signal levels, MLT-3 uses 3 levels: $+V$, 0 , $-V$
- *Complex* transition rules
 - If next bit = 0 , no transition
 - If next bit = 1 and current level = $\pm V$, transition to level 0
 - If next bit = 1 and current level = 0 , transition to opposite of last nonzero level

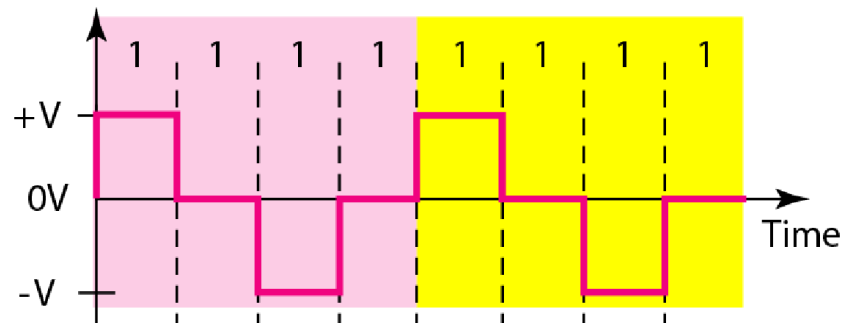
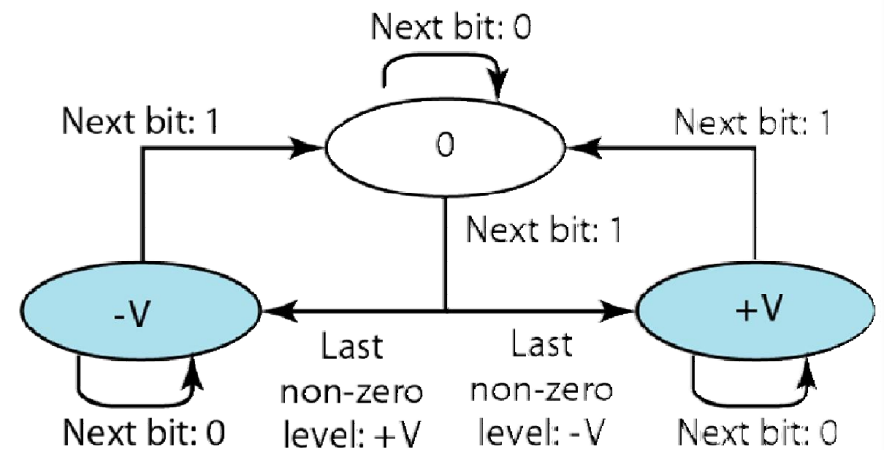
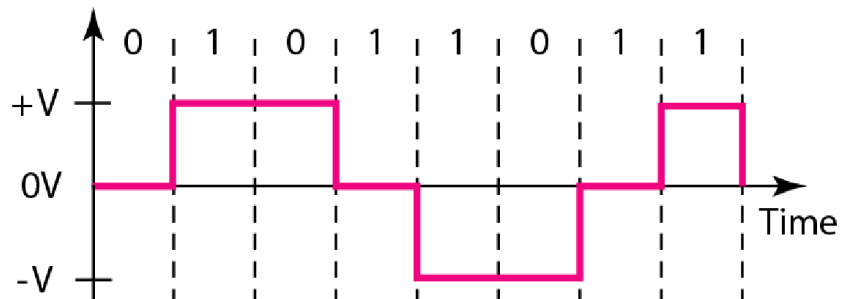


Multitransition: MLT-3



- *Complex* transition rules
- $r = 1$
- No synchronization for long sequence bit 0's

Multitransition: MLT-3



Nonperiodic to periodic signal
with a period of 4× bit duration

Repeating pattern: +V0-V0

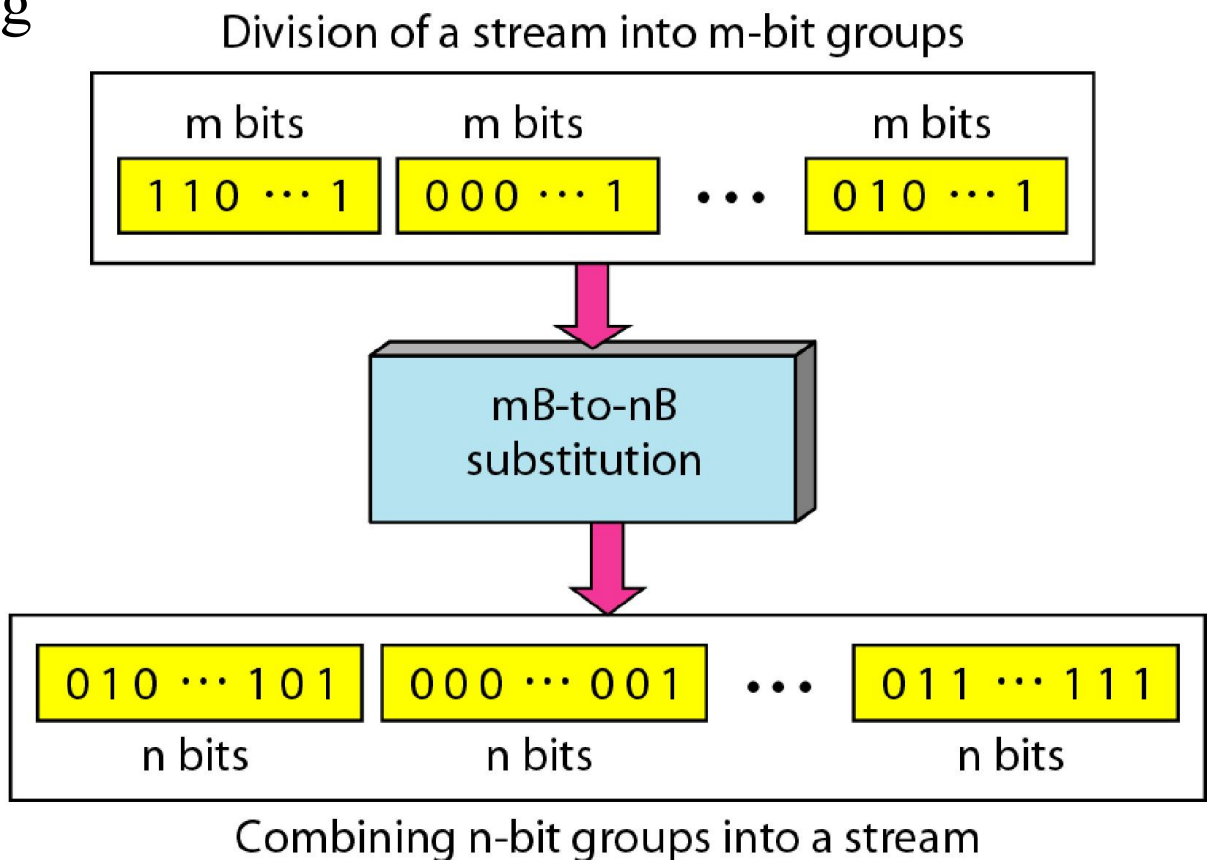
Line Coding summary

<i>Category</i>	<i>Scheme</i>	<i>Bandwidth (average)</i>	<i>Characteristics</i>
Unipolar	NRZ	$B = N/2$	Costly, no self-synchronization if long 0s or 1s, DC
Polar	NRZ-L	$B = N/2$	No self-synchronization if long 0s or 1s, DC
	NRZ-I	$B = N/2$	No self-synchronization for long 0s, DC
	Biphase	$B = N$	Self-synchronization, no DC, high bandwidth
Bipolar	AMI	$B = N/2$	No self-synchronization for long 0s, DC
Multilevel	2B1Q	$B = N/4$	No self-synchronization for long same double bits
	8B6T	$B = 3N/4$	Self-synchronization, no DC
	4D-PAM5	$B = N/8$	Self-synchronization, no DC
Multiline	MLT-3	$B = N/3$	No self-synchronization for long 0s

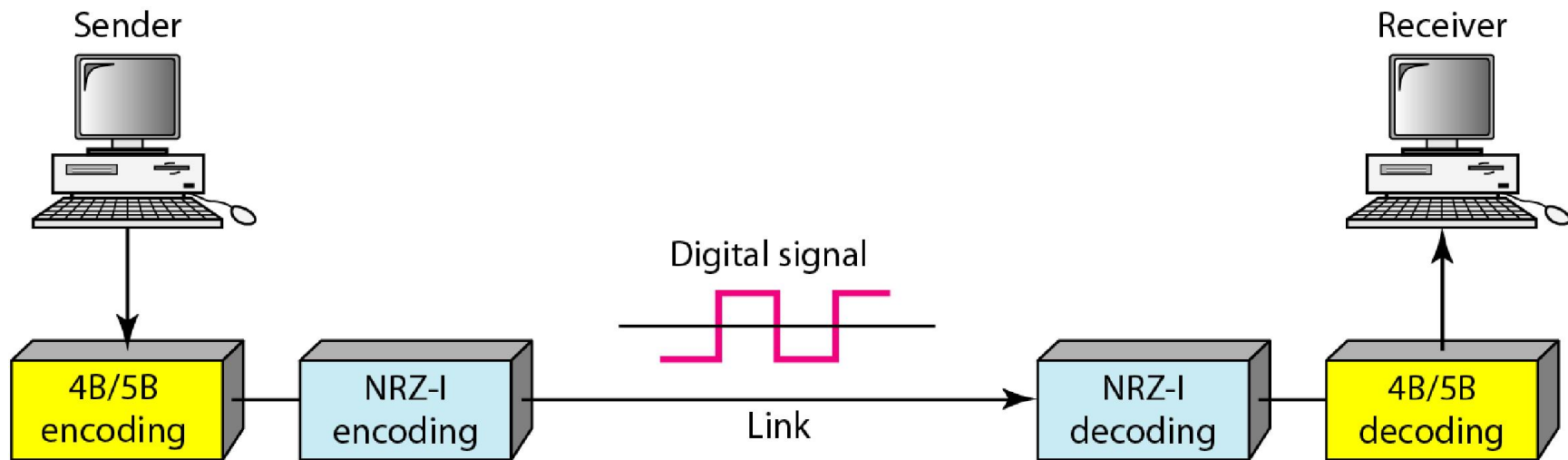
Block Coding: mB/nB

- replaces m bits by n bits, where $m < n$
- increases redundancy and synchronization
- Done before line coding using 3 steps

- division
- substitution
- combination



Block Coding: *4B/5B*

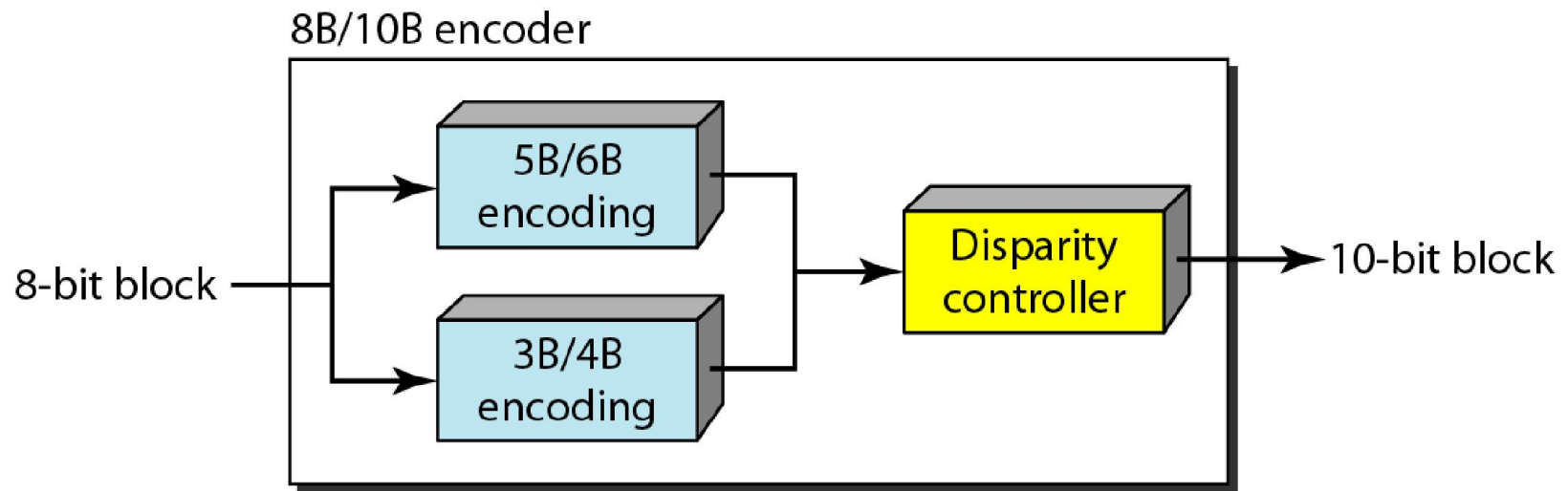


- Done in combination with NRZ-I which has sync problem for long sequence of 0's
- *4B/5B* ensures no more than 3 consecutive 0's

Block Coding: 4B/5B

<i>Data Sequence</i>	<i>Encoded Sequence</i>	<i>Control Sequence</i>	<i>Encoded Sequence</i>
0000	11110	Q (Quiet)	00000
0001	01001	I (Idle)	11111
0010	10100	H (Halt)	00100
0011	10101	J (Start delimiter)	11000
0100	01010	K (Start delimiter)	10001
0110	01110	S (Set)	11001
0111	01111	R (Reset)	00111
1000	10010	16 data sequences 32 available encoding sequences - redundant sequences are for error detection, overhead control - Maximum 1 leading 0 and 2 trailing 0's	
1001	10011		
1010	10110		
1011	10111		
1100	11010		
1101	11011		
1110	11100		
1111	11101		

Block Coding: $8B/10B$



- More redundancy and better error detection than $4B/5B$
- Actually done by combination of $5B/6B$ and $3B/4B$ to simplify mapping table

Search for Best Line Coding: Scrambling

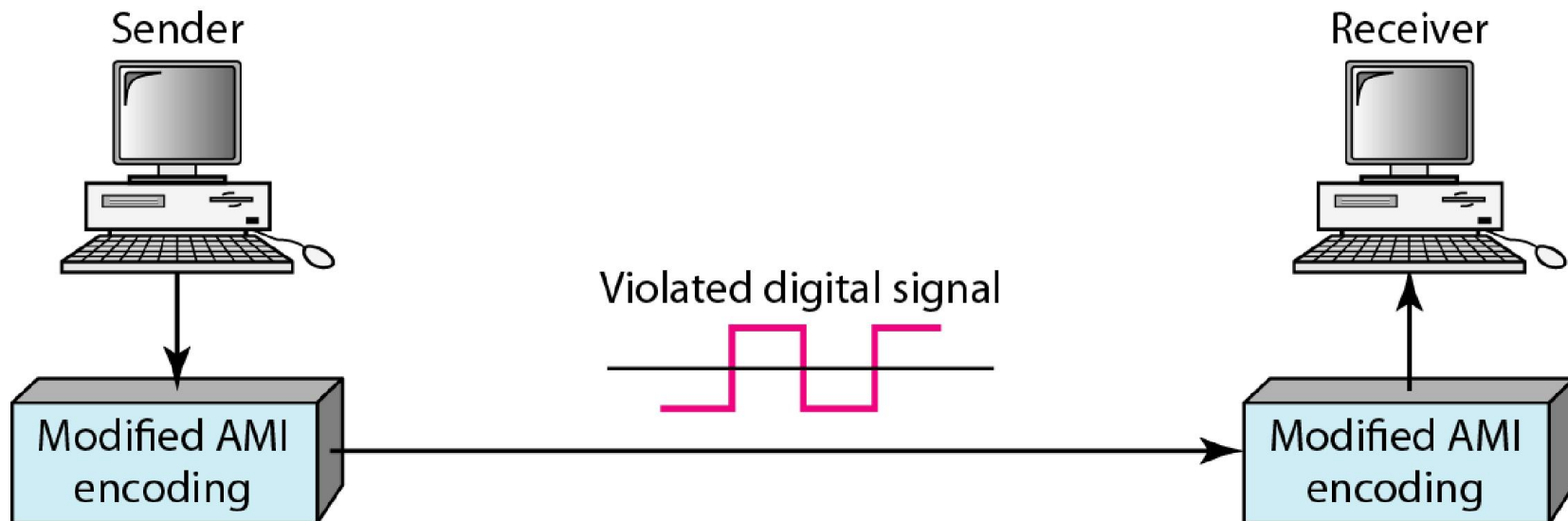
- Characteristics of a good line coding
 - No increase in b/w for self sync
 - NO DC
- Biphasic scheme
 - High bandwidth ($r < 1$)
- NRZ + block coding
 - DC still exists
- Bipolar AMI
 - NO sync for long sequence of 0's
 - Other characteristics are OK

Search for best line coding: Scrambling

- Bipolar AMI
 - NO sync for long sequence of 0's
 - Other characteristics are OK
- Scrambling uses AMI but inserts nonzero pulses in long sequence of 0's

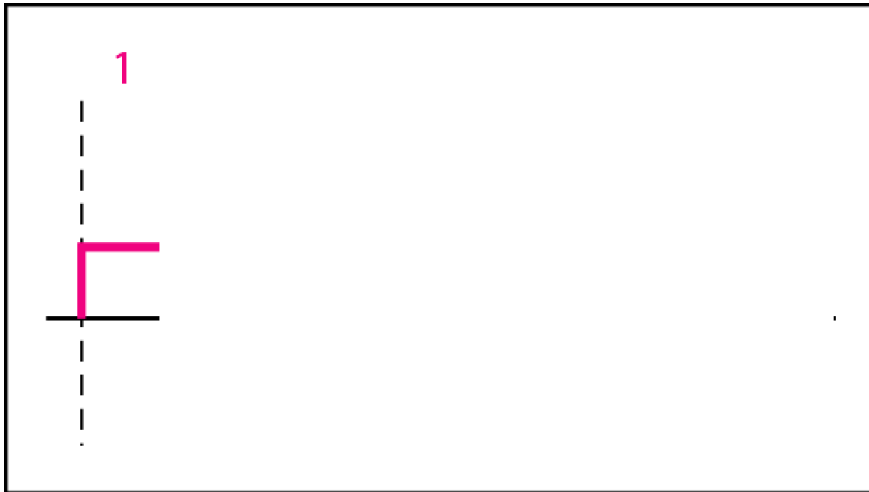
Search for best line coding: Scrambling

- Bipolar AMI
 - NO sync for long sequence of 0's
 - Other characteristics are OK
- Scrambling uses AMI but inserts nonzero pulses in long sequence of 0's
- Thus AMI's rules are violated on the fly



Scrambling: *B8ZS*

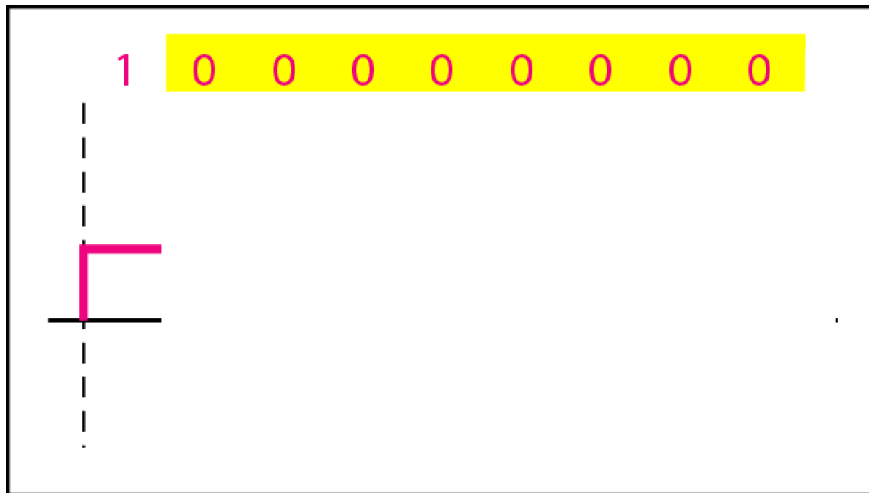
- Bipolar 8 Zeros (*B8ZS*)
 - Replaces 8 zeros by 0 0 0 V B 0 V B
 - V means Violation to AMI rules
 - B means normal polarity according to AMI rules



a. Previous level is positive.

Scrambling: *B8ZS*

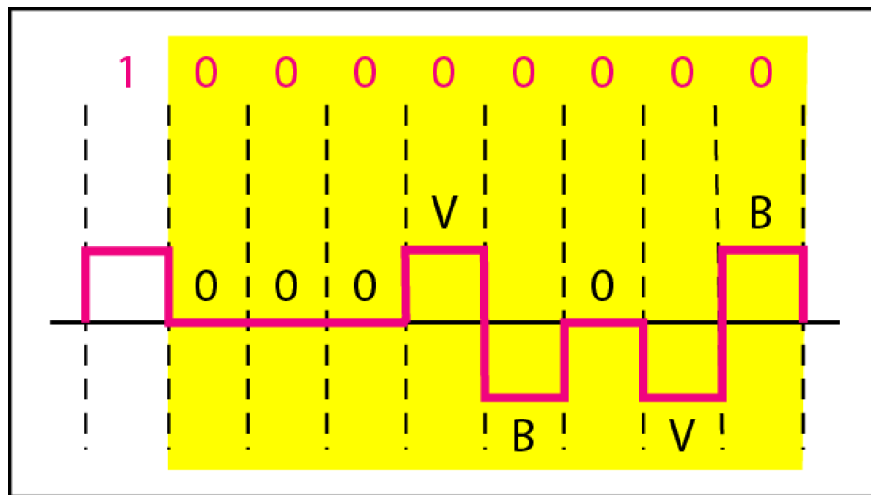
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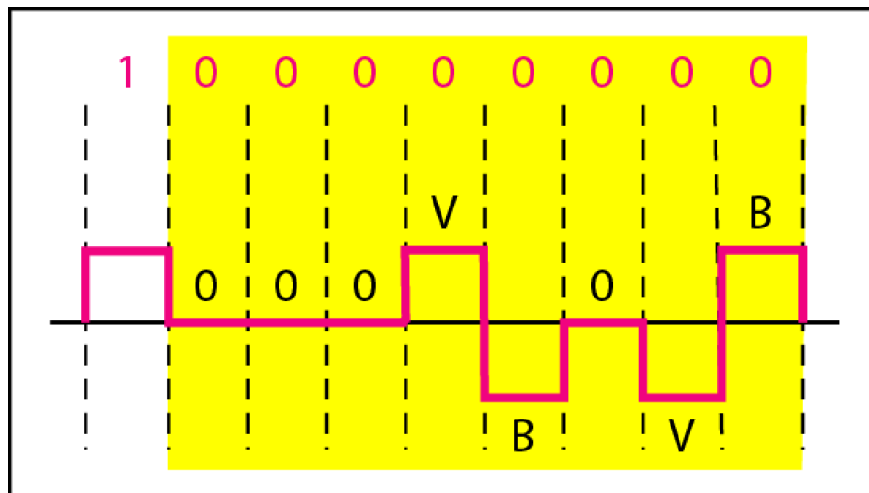
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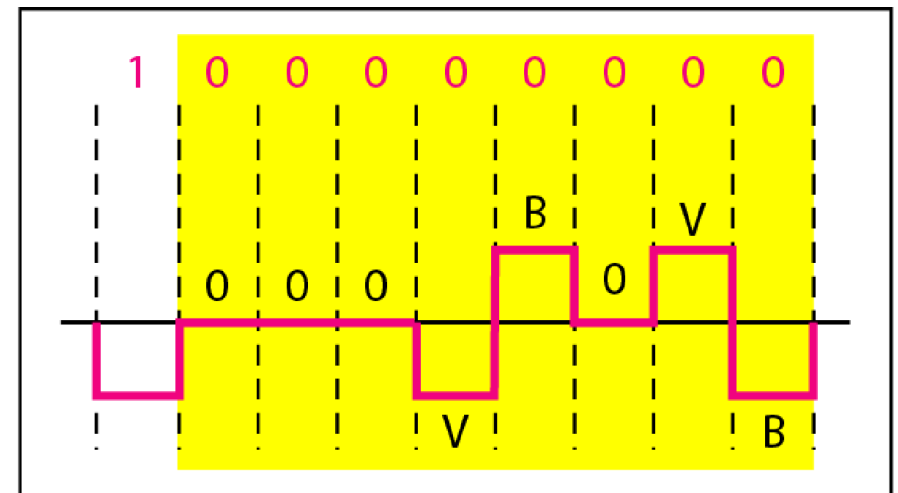
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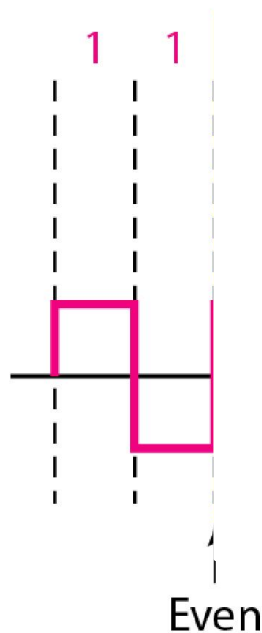
b. Previous level is negative.

Scrambling: *HDB3*

- High Definition Bipolar 3-Zero (*HDB3*)
- Replaces 4 zeros by
 - 0 0 0 V if No. of nonzero pulses after last substitution is ODD
 - B 0 0 V if No. of nonzero pulses after last substitution is EVEN

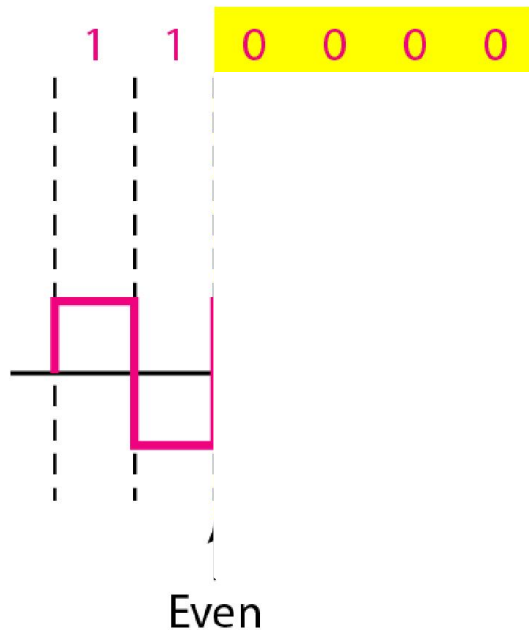
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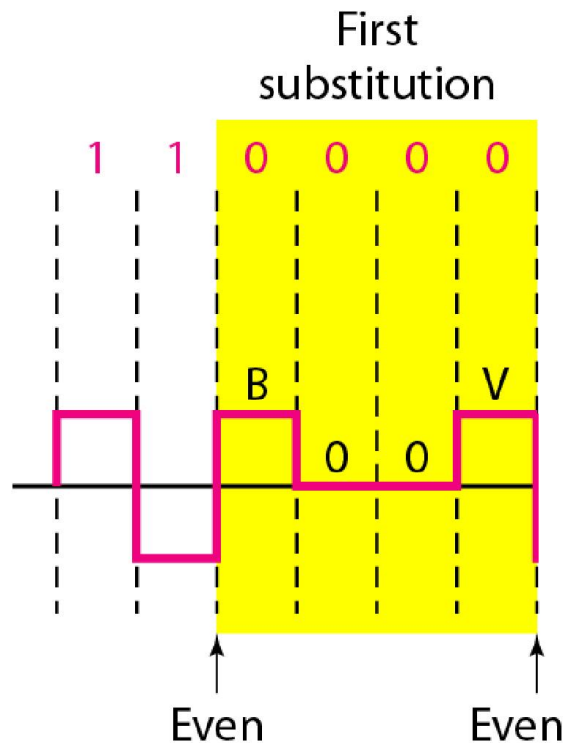
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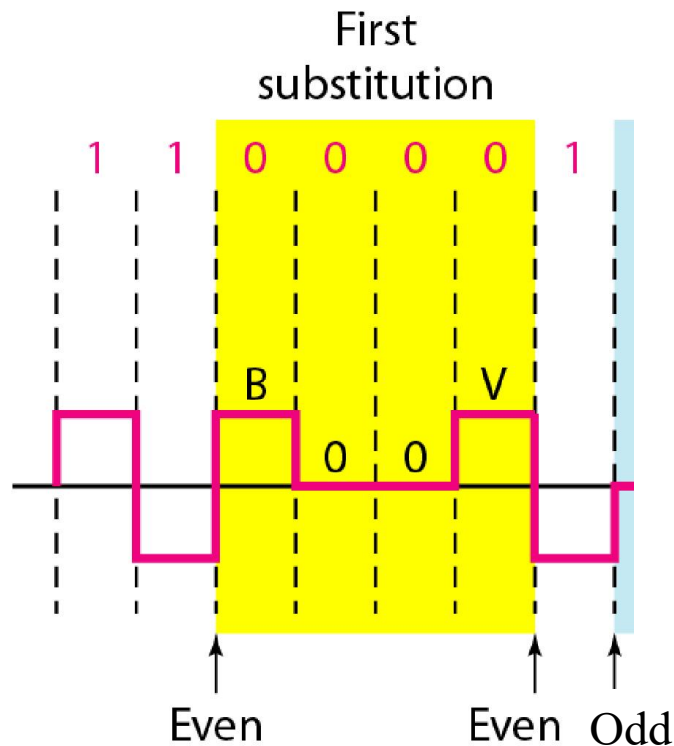
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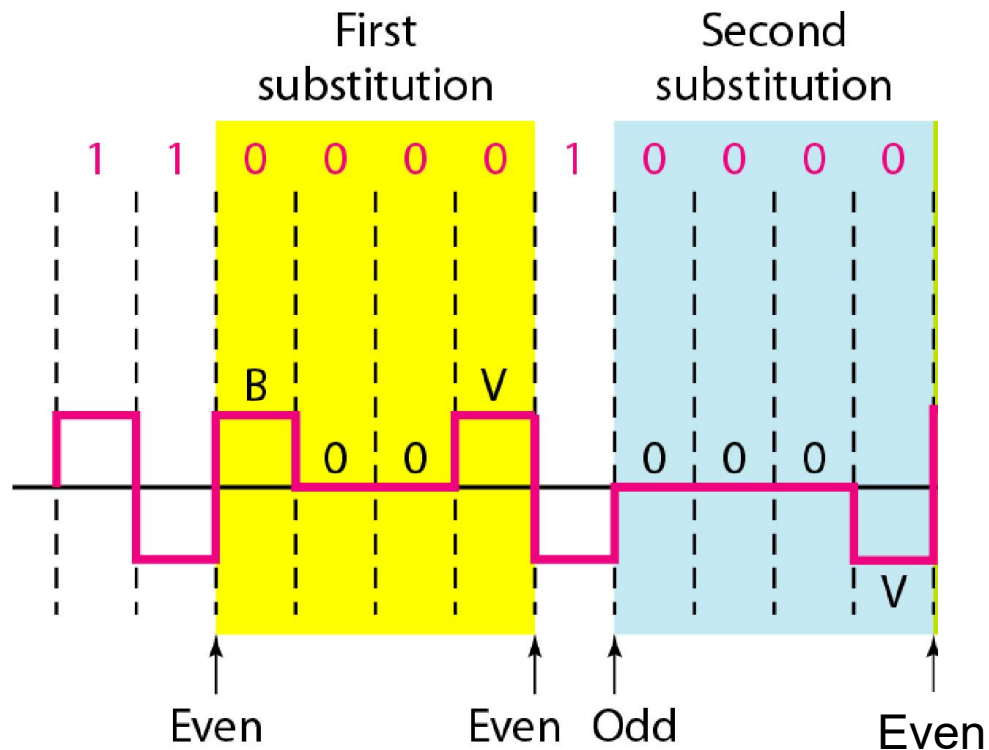
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